St. PETER'S ENGINEERING COLLEGE
Giving Wings To Thoughts
(UGC - AUTONOMOUS | Affiliated to JNTU Hyderabad)
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Department of Computer Science and Engineering –AIML
Project Stage I - First Review - Two page Write-Up

Tittle of the Project: NUTRI LENSE AI

Domain : Nutri Lens AI is an AI-powered HealthTech solution (CSE-AIML) that utilizes Tesseract OCR and Python to extract ingredient text from packaged food images, analyze health risks against personalized user profiles (such as medical conditions or allergies), and recommend healthier dietary alternatives

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Problem Statement: Consumers struggle to interpret complex packaged food labels due to technical ingredient names and food additives, while standard nutrition labels fail to provide personalized guidance tailored to individual medical conditions (such as diabetes or high blood pressure), allergies, or fitness goals

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Proposed Approach: Nutri Lens AI proposes an automated, AI-driven solution that simplifies nutritional label analysis by combining Tesseract Optical Character Recognition (OCR), artificial intelligence, and a structured nutrition database. The system allows users to upload an image of a packaged food label, automatically extracts the raw ingredient text, and scans for harmful additives, preservatives, and potential allergens. It then evaluates these ingredients against the user's registered health profile—factoring in age, fitness goals, dietary preferences, and medical conditions such as diabetes, high blood pressure, or food allergies—to calculate a personalized health rating. Finally, the platform delivers easy-to-understand explanations for its evaluation and recommends healthier product alternatives to help consumers make informed eating choices

Key Modules:

- User Profile Management: Stores user health details, allergies, and medical conditions.
- OCR Label Scanning: Extracts raw ingredient text from product photos using Tesseract OCR.
- AI Ingredient Analysis: Identifies harmful additives, preservatives, and potential allergens.
- Personalized Health Evaluation: Evaluates extracted ingredients against the user's health profile to generate a health rating.
- Product Recommendation Engine: Suggests healthier alternative food products

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System Requirements

- Hardware : intel i3 (min) , 8 GB RAM (min) , 512 GB hard disk
- Operating System : Windows 10 (min)
- Coding Language : Python 3.13 , HTML , CSS
- Frame Work : Flask 3.0
- Data Base : MySql 5.5

Team Details:

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Project Guide **HOD CSE(AIML)**

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